IPM-Net: An Interpretable Partial-Level Matching Network for Few-Shot Fine-Grained Classification in Optical Remote Sensing Images

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***Abstract*—Deep learning has enabled AI-driven fine-grained object classification in optical remote sensing imagery with remarkable precision. However, insufficient training samples critically hinder their performance in data-scarce scenarios such as military operations and natural disaster response, significantly diminishing the practical utility of deep learning-based approaches. Consequently, few-shot fine-grained classification methods have garnered substantial attention due to their superior ability in knowledge learning and transfer under limited data conditions. Current approaches prioritize discriminative feature extraction and enhancement, but lack mathematical definition of fine-grained features and an explainable analysis of how these features influence decision-making processes. This deficiency prevents models from explicitly identifying critical features for knowledge transfer. In response to this limitation, we introduce the Interpretable Partial-level Matching Network (IPM-Net), the main goal is to match the query target's partials in the support set, thus, building and visualizing the decision-making process. This network consists of three modules: a Partial-level Game-theoretic Interpretable (PGI) Module, Cross-pose Semantic Alignment (CSA) Module, and Refine-and-Rediscover (RaR) Module. Experiments on a variety of fine-grained datasets—both remote sensing and natural images—confirm IPM-Net's superiority in accuracy and interpretability. The source code is publicly available on GitHub: https://github.com/Silver2kw/IPM-Net**

***Index Terms*—Deep learning, few-shot learning, fine-grained classification, interpretability, remote sensing images.**

I. Introduction

FINE-GRAINED classification is a key task in the intelligent interpretation of remote sensing images, focusing on distinguishing subtle differences among objects within the same basic category [1], [2]. It plays a critical role in applications such as airspace control, situational assessment, and monitoring of critical port targets [3]. While optical remote sensing image fine-grained classification has achieved remarkable accuracy, insufficient training samples hindering adequate learning of interclass differences in discriminative features remains a critical bottleneck in data-scarce scenarios [4]. To address this challenge, few-shot fine-grained classification has spurred significant research efforts to achieve high-precision classification under constrained optical remote sensing data conditions [5]. The core objective lies in distinguishing targets with visually similar partials through limited samples.

Current methodologies focus on extracting, preserving, and enhancing critical fine-grained essential features from support sets. The core approaches can be summarized into two key strategies, one focuses on extracting and enhancing fine-grained features. Pioneering work by Lin et al. [6] introduced bilinear CNNs to model interactions between object regions, establishing a foundation for local feature representation, and for instance, Gidaris et al. [7] designed rotation-prediction-driven attention to emphasize semantically meaningful areas, while Lee et al. [8] further enhanced task adaptability through task-discrepancy maximization. The second paradigm emphasizes knowledge transfer mechanisms, addressing the generalization challenges in few-shot scenarios by redefining cross-task feature interactions; a seminal contribution by Doersch et al. [9] proposed CrossTransformers, which mitigates supervision collapse by establishing spatial correspondences between support and query images, enabling robust cross-domain alignment, and for remote sensing applications, RIFRN [10] devised rotation-invariant modules with complementary error metrics to address orientation variations, collectively advancing feature reuse across tasks while maintaining sensitivity to fine-grained distinctions under limited supervision.

While current methods aim to identify subtle interclass distinctions through enhancing fine-grained features and data transferring, they lack mathematically defining fine-grained feature semantics, nor transparently trace knowledge transfer from support to query sets. As depicted in Fig. 1, existing methods execute both feature extraction and matching processes as black boxes, with their underlying decision-making mechanisms remaining opaque. Lundberg and Lee [11]

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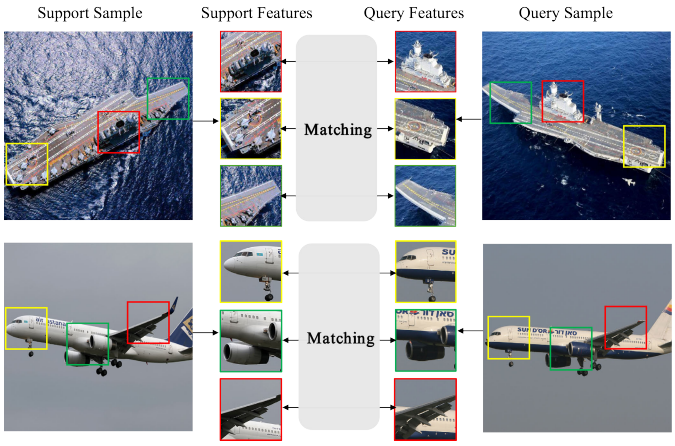


Fig. 1. Current fine-grained decision mechanism in few-shot learning, where features are extracted from support and query samples and then matched for detailed classification.

proposed Shapley values for feature contribution analysis, upon which Cheng et al. [12] constructed a cooperative game-theoretic framework to quantify the importance of individual tokens in language models. This theoretical foundation enables rigorous mathematical definition of game-theoretic interactions among fine-grained partial components in remote sensing imagery. Besides, saliency map technologies such as Grad-CAM [13] has emerged as a prevalent visualization technique, which can be applied to effectively demonstrate how knowledge acquired from support sets informs decision-making processes for query samples. By leveraging interpretability as a strategic mechanism, encompass mathematical formalization of partial component interactions to identify essential characteristics critical for fine-grained categorization, and visualization-driven optimization of feature matching processes based on knowledge transfer partials.

To overcome the inherent limitations in current methods—specifically, the lack of clear semantic interpretation in the extracted features and the ambiguity of knowledge transfer paths—we propose Interpretable Partial-level Matching Network (IPM-Net), a novel framework that focuses on matching partial features between query targets and the support set to establish an interpretable decision-making process. This approach integrates three complementary modules to address these challenges. The first component, the Partial-Level Game-theoretic Interpretable (PGI) Module, facilitates the extraction and visualization of partial features from both support and query sets. We leverage a learnable attention mechanism to model pixel-level relationships, combined with a dynamic update process driven by LSTM, while optimizing the system with a combined loss function that enforces both intra-aggregation and component-separation constraints. The second module, the Cross-pose Semantic Alignment (CSA) Module, effectively reduces posture and scale discrepancies between the support and query sets, thereby clarifying the knowledge transfer path at the semantic level. Through a partial-level similarity matrix, we align the features semantically, ensuring robust matching despite variations in object pose and scale. The final module, Refine-and-Rediscover (RaR) Module, improves both the efficiency of feature extraction and the generalization ability of the model. The refine branch selectively generates masks to crop the main target area, reducing background noise, while the rediscover branch identifies discarded samples, guiding the network to explore additional feature domains and enhancing its robustness.

Our contributions can be summarized as follows.

1) Proposed a cooperative game theory between partial-level features for mathematical modeling few-shot fine-grained classification in remote sensing images.
2) As a white-box-oriented model with intrinsic interpretability, we introduce a PGI Module, which embeds explanation directly into the decision-making process by modeling partial-level features as a cooperative game. This design promotes the decision-making process and enables the visualization of knowledge transfer in a human-understandable manner.
3) Established a CSA Module to align object morphology at the semantic level and optimize feature representations of deformed query-support pairs, specifically designed to address variations in object postures across samples.
4) Designed a RaR Module based on the cooperative game between foreground and background features, which adaptively refines the target image by focusing on the most discriminative regions.

The remainder of the article is organized as follows. We describe related works in Section II and introduce our proposed IPM-Net in Section III. Evaluations on two public datasets and our own dataset are presented in Section IV, followed by conclusions in Section V.

II. Related Works

A. Few-Shot Fine-Grained Classification

Few-shot fine-grained image classification (FS-FGIC) seeks to recognize visually similar categories from limited labeled samples, such as distinguishing bird species with subtle plumage variations or identifying car models that differ only in local structural details. Compared to conventional few-shot learning, FS-FGIC demands stronger discriminative representation and cross-task generalization abilities. Existing studies can be broadly categorized into two directions: feature enhancement, which aims to learn and organize more discriminative representations, and knowledge transfer, which focuses on improving generalization across base and novel classes.

1) Feature Enhancement: Feature enhancement in FS-FGIC primarily addresses two questions: 1) *what to learn*—how to extract sufficiently discriminative fine-grained features from limited samples and 2) *how to organize*—how to integrate multilevel representations without introducing redundancy or noise.

a) Discriminative feature extraction: The key challenge in fine-grained classification lies in identifying the most informative and discriminative regions. Wei et al. [14] formalized the FS-FGIC task and introduced bilinear feature models to learn piecewise mapping classifiers that capture interactions

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among local regions. Following this line, Zhu et al. [15] proposed MattML, an attention-based task embedding framework that enables task-specific initialization. Xu et al. [16] integrated spatial and channel attention to extract complementary information, where spatial attention captures localized cues and channel attention encodes holistic semantics. Transformer-based architectures such as HelixFormer [17] extend this concept by employing cross-image attention to enhance the model's ability to detect subtle discriminative patterns. Wu et al. [18] propose a trilinear spatial-awareness module that simultaneously mines selective, structural, and subtle patterns. Recent work has further emphasized foreground-aware feature reconstruction to mitigate background dominance in few-shot scenarios [19]. Collectively, these studies emphasize discriminative feature extraction through spatial focus, channel selection, and part-aware modeling.

b) Multilevel feature fusion: Beyond discriminative feature extraction, an essential question concerns how to effectively organize features across multiple network layers. Direct aggregation of mid-level features often introduces background noise, leading to potential overfitting in few-shot scenarios. To mitigate this, BSFA [20] adopts a two-stage framework combining background suppression and foreground alignment to improve sample correspondence. Other approaches, such as bidirectional reconstruction networks [21], further refine feature representations by balancing interclass and intraclass variations. Wu et al. [18] introduce Semantic-guided Generalization Enhancement (SGE) that learns a lightweight gating function to reweight multilevel features according to their estimated consistency with support layers. Similarly, Gu et al. [22] propose a multistage fusion of local and global features that progressively aggregates stage-wise embeddings via a learnable calibration factor. These methods collectively demonstrate that effective multilevel feature fusion and noise suppression are critical in enhancing model robustness under limited supervision.

2) Knowledge Transfer: While feature enhancement improves representational discriminability, it alone is insufficient for robust generalization to unseen categories. Knowledge transfer methods address this limitation by enabling the model to generalize across tasks, effectively learning transferable meta-knowledge rather than memorizing task-specific patterns. Knowledge transfer approaches can be broadly categorized into reconstruction-based, alignment-based, and prompting-based methods. Reconstruction-based frameworks, such as FRN [23] and BiFRN [21], reformulate classification as a reconstruction problem, learning to recover query features from support samples to enhance interclass separability. Alignment-based methods, including OLSA [24] and CrossTransformers [9], explicitly align spatial or semantic correspondences between query and support features, thereby improving cross-task and cross-domain consistency. More recent prompting-based frameworks introduce transferable meta-knowledge into few-shot models, these methods enable consistent adaptation to novel categories without extensive retraining [25].

B. Explainable Artificial Intelligence

Explainable Artificial Intelligence (XAI) aims to enhance the transparency and understanding of artificial intelligence systems. Within this context, explainability and interpretability are closely related but conceptually distinct. Explainability refers to the ability to provide external, posthoc explanations for an already-trained model, whereas interpretability denotes the property of models whose internal decision logic is intrinsically comprehensible to humans [26]. Barredo Arrieta et al. [27] further clarify that interpretability is a structural characteristic of the model itself, while explainability is an auxiliary function that can be attached to otherwise opaque systems. Based on where and how explanatory information is introduced, existing XAI methods are commonly categorized into three groups: premodel, in-model, and postmodel explanations. This taxonomy, formally proposed by Tjoa and Guan [28] and later adopted in multiple surveys, reflects increasing levels of model dependency.

1) Premodel: Premodel explanations focus on constructing models or representations that are interpretable by design. These methods emphasize interpretability by constraining model complexity or adopting transparent formulations. Representative approaches include interpretable prediction models developed for high-stakes domains [29]. Game-theoretic formulations, particularly those based on Shapley values, are also widely adopted to characterize feature contributions in a principled and human-understandable manner [30].

2) In-Model: In-model explanations integrate explanatory mechanisms directly into the learning process, occupying a middle ground between interpretability and explainability. Typical examples include attention mechanisms and saliency-based modules, which expose intermediate decision cues while preserving model flexibility. In few-shot fine-grained classification, Zarei and Komeili [31] proposed concept-based prototypical networks that incorporate human-interpretable concepts into the model structure. Norrenbrock et al. [32] introduced Sparse Low-Dimensional Decision Models to restrict decision-making to a compact set of informative features. Xue et al. [33] developed the Region Comparison Network, which visualizes discriminative regions to facilitate understanding of fine-grained recognition behavior. These methods improve transparency by partially exposing internal reasoning, though their explanations remain model-dependent.

3) Postmodel: Postmodel explanations aim at explainability by analyzing trained models without modifying their internal structures. Such post hoc methods are particularly suited to complex black-box models. In remote sensing image classification, six major categories of postmodel explainability techniques have been summarized, including Activation Maximization [34], Surrogate Models [35], [36], Attribution Methods [37], Perturbation-based Methods [38], Class Activation Maps [39], and Example-based Methods [40], [41]. While effective for optical remote sensing images, these approaches often struggle with large-scale scenes due to limited scalability and contextual consistency [42]. Recently, foundation models like SAM have shown promise in addressing limited data challenges in remote sensing tasks such as change detection [43]. To address this issue,

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Zhu et al. [44] incorporated prior knowledge and geological abstraction rules to propose Hybrid-Grid, an explainable and scale-adaptive framework-consistent method.

III. Methodology

A. Mathematical Modeling

Existing image classification tasks are typically modeled as a mapping from input images to output labels. However, for the target classification task in high-resolution remote sensing imagery under complex scenes, it is necessary to construct an explainable decision-making probability model based on discriminative features and the structure of the classification model, as illustrated in the following equation:

$$p_{\text{trustworthy}} = g_\theta\left(f_{\text{ess}}, \mathcal{M}\right) \tag{1}$$

where $p_{\text{trustworthy}}$ indicates the decision-making probability of the object classification task, $g(cdots)$ is the process of explainable classification, θ is the parameters of the model, f_{ess} is the discriminative features that the classification task relies on, and $\mathcal{M}$ is the structure of the classification model.

Due to the characteristics of limited sample size and small interclass differences in few-shot fine-grained classification tasks, the classification model needs to focus on the discriminative features who describing the differences between similar subclasses, that is, the essential feature to determine which subclass that the object belongs to. In order to improve the accuracy and interpretability of few-shot fine-grained object classification, it is necessary to finely modeling the essential features of the object. The essential features are defined as follows:

$$f_{\text{ess}} = \alpha f_{\text{obj}} + (1-\alpha) f_{\text{background}} + \beta f_{\text{interf}} \tag{2a}$$

$$= \alpha \sum_m f^m_{\text{partial}} + (1-\alpha) f_{\text{background}} + \beta f_{\text{interf}} \tag{2b}$$

where f_{ess} indicates the essential feature, f_{obj} is the intrinsic feature of object, $f_{\text{background}}$ is the background feature of object, the effective feature consists of f_{obj} and $f_{\text{background}}$, while f_{interf} is disturbing features. f_{partial} is the partial features who construct the object. $\alpha \in [0,1]$ is the factor of object feature, as well as $\beta \in [0,1]$ is the factor of disturbing features.

In the process of feature extraction in the model, there is a specific mapping relationship between each component of the essential features, the feature map elements, and the pixels of input image. For few-shot fine-grained object classification in remote sensing images, it is necessary to utilize semantic entities with clear boundaries to describe the object. Therefore, by examining the contribution of each pixel in the image at microscale, with each pixel in the spatial semantic entity as the minimum analysis unit, object, partial, and background can be conclude are as following pixel sets:

$$\Psi^m_{\text{partial}} = \{e_i\}^{k_m}_{i=1} \tag{3}$$

$$\Psi_{\text{obj}} = \left\{\Psi^m_{\text{partial}}\right\}^t_{m=1} \tag{4}$$

$$\Psi_{\text{background}} = \left\{e_j\right\}^{k_{\text{bg}}}_{j=1}, \quad j \neq i. \tag{5}$$

Among them, e is a feature map element (or image pixel) that has a clear mapping relationship with the position of the input image, Ψ_{partial} represents the set of elements for the mth component of the target, and k_m is the total number of elements for the mth component. Ψ_{obj} represents the set of target elements, which includes all the components in the target. t is the total number of components in the target. $\Psi_{\text{background}}$ represents a set of background elements that only contain background information of the feature map (or original image). k_{bg} represents the number of elements in the background set.

Based on the feature stream in the model, we can define: 1) for the initial input image, Ψ is the set of all pixels participating in the game and 2) for the feature map output during the classification process, Ψ is the set of all elements in this feature map. Therefore, for the minimum analyzed pixel e, its Shapley Value $\mathcal{S}(e;\Psi)$ is

$$\mathcal{S}\left(e;\Psi\right) = \sum_{\Omega\subseteq\Psi\setminus\{e\}} \frac{(|\Psi|-|\Omega|-1)!|\Omega|!}{|\Psi|!}\left[v\left(\Omega\cup\{e\}\right)-v\left(\Omega\right)\right] \tag{6a}$$

$$= E_{\Omega\subseteq\Psi\setminus\{e\}}\left[v\left(\Omega\cup\{e\}\right)-v\left(\Omega\right)\right]. \tag{6b}$$

Among them, $|\cdot|$ represents the total number of elements in the set, Ω is any possible subset of Ψ, $v(\Omega)$ denotes the output (probability) of subset Ω in the classification, $\mathcal{S}$ is the Shapley value, and E indicates taking the expectation over all possible subsets.

To accurately describe the cooperative game relationship, we design the contribution game and influence game models to evaluate the strength and correlation of the two objects participating in the game.

The contribution game focuses on the benefit distribution of the two participants U_1 and U_2. If $\mathcal{G}_1(U_1,U_2) > 0$, then U_1's contribution is greater than U_2; if $\mathcal{G}_1(U_1,U_2) = 0$, then U_1's contribution is equally important as U_2; if $\mathcal{G}_1(U_1,U_2) < 0$, then U_2's contribution is greater

$$\mathcal{G}_1\left(U_1,U_2\right) \overset{\text{def}}{=} \sum_{i=1}^{|U_1|}\mathcal{S}\left(e_i;U_1\right) - \sum_{j=1}^{|U_2|}\mathcal{S}\left(e_j;U_2\right). \tag{7}$$

The influence game focuses on the impact of the presence of U_2 on U_1. If $\mathcal{G}_2(U_1,U_2) > 0$, then the presence of U_2 has a positive impact on U_1's contribution to the game; if $\mathcal{G}_2(U_1,U_2) = 0$, then the presence of U_2 has no impact on U_1's contribution to the game; if $\mathcal{G}_2(U_1,U_2) < 0$, then the presence of U_2 has a negative impact on U_1's contribution to the game

$$\mathcal{G}_2\left(U_1,U_2\right) \overset{\text{def}}{=} \sum_{i=1}^{|U_1\cup U_2|}\mathcal{S}\left(e_i;U_1\cup U_2\right) - \sum_{j=1}^{|U_1|}\mathcal{S}\left(e_j;U_1\right). \tag{8}$$

B. Architecture Overview

Building upon the mathematical formulation, the theoretical foundation described above guides the structural design of IPM-Net. The overall architecture of IPM-Net is illustrated in Fig. 2. It provides a visual overview of the network's dual-stage design, consisting of the PGI Module, the CSA Module, and the RaR Module.

The PGI module (detailed in Section III-C) formulates a game-theoretic interaction among feature subregions to

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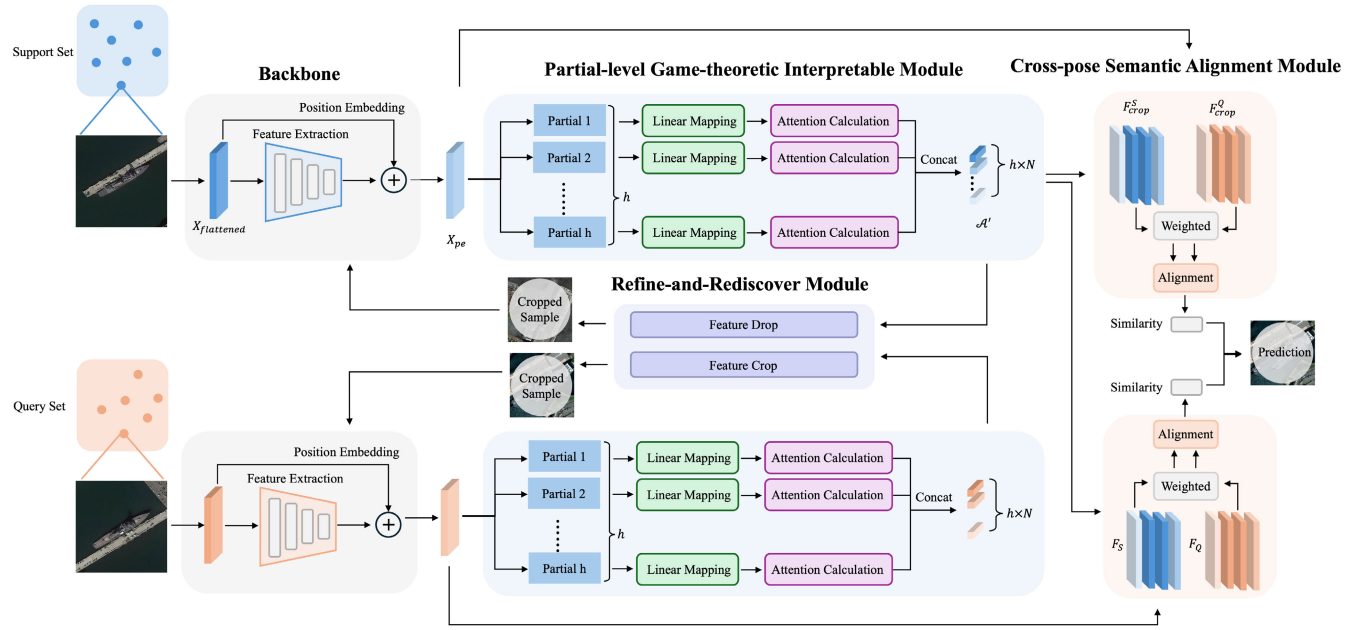


Fig. 2. Overview of the IPM-Net. For each support-query sample pair, the process begins with the backbone network. Subsequently, the PGI module is employed to extract partial-level features. These features are then fed into the cross-pose semantic alignment module to achieve semantic alignment between the support and query samples. Concurrently, the partial-level features are input into the RaR module, that generates a cropped sample. The cropped support-query sample pair is then reprocessed through the backbone network and the Cross-pose Semantic Alignment Module to obtain the semantically aligned features of the cropped support and query. For both the original and cropped samples, the similarity between the query-support aligned image pairs is calculated. The final decision is supported by integrating the similarities of these two image pairs.

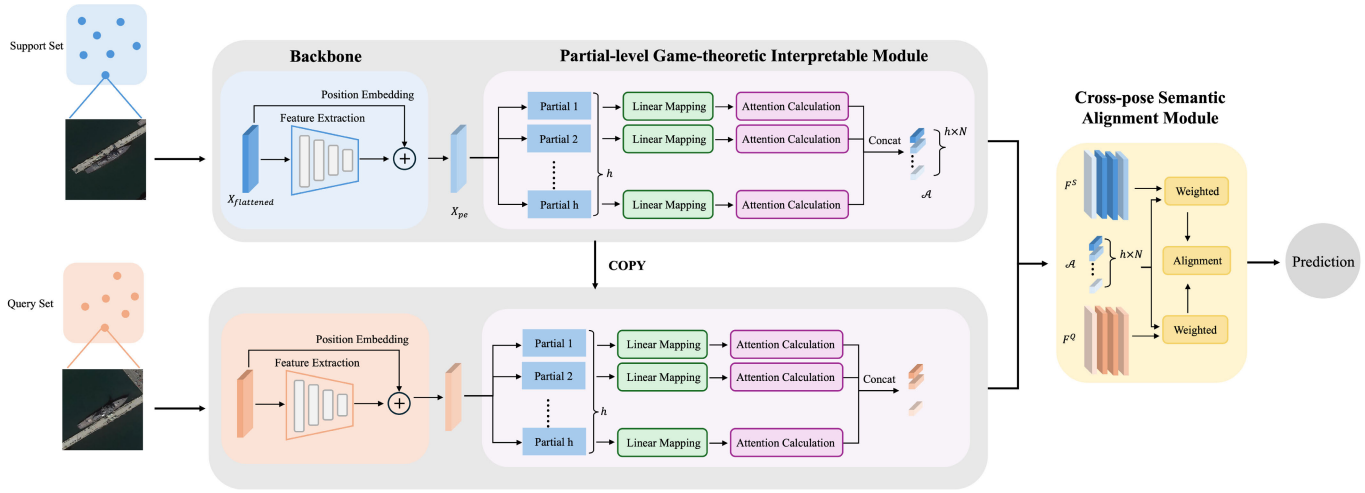


Fig. 3. Overall inference pipeline of the proposed IPM-Net. During inference, both the query and support images are processed through the shared backbone and PGI module to extract partial-level features. The CSA module then aligns the corresponding features across different poses and scales to compute semantic similarity, while the RaR module is disabled during inference to ensure efficient and deterministic prediction. The final classification decision is obtained by fusing the similarity scores derived from the aligned features.

adaptively decompose complex remote sensing targets into semantically meaningful partials. This process enables interpretable componentwise reasoning and encourages the network to capture fine-grained structures beyond holistic representations. The CSA module (discussed in Section III-D) enhances the discriminative capacity of these partials by establishing semantic correspondence between query and support features. Through cross-pose alignment, CSA effectively mitigates geometric and viewpoint inconsistencies, ensuring that the semantically aligned partials learned by PGI maintain consistent meaning across different spatial configurations. In this way, CSA acts as a bridge between interpretable part reasoning and semantic transfer, reinforcing PGI's capability in cross-instance alignment. Finally, the RaR module (explained in Section III-E) operates as a self-refining mechanism that dynamically focuses on the most informative object regions while rediscovering complementary cues neglected in the initial reasoning stage. By iteratively refining and rediscovering regions, RaR improves both the robustness and generalization of the learned representations, effectively suppressing irrelevant background information and enhancing target saliency.

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For the training pipeline of the model, both the original stage and the RaR stage share the same feature extractor and PGI parameters, while maintaining independent global classifiers to adapt to the distinct feature distributions of each stage. The RaR module operates exclusively during training, generating refined images that guide the network to focus on discriminative target regions and to suppress background interference. More training details will be elaborated in Section III-F. Furthermore, the overall inference pipeline of the proposed method is depicted in Fig. 3. During inference, only the backbone, PGI, and CSA modules are employed. It illustrates how the model processes query and support samples through shared feature extraction, interpretable partial decomposition, semantic alignment, and final similarity-based decision fusion.

In summary, IPM-Net integrates PGI for interpretable partial-level feature decomposition, CSA for semantic alignment across poses, and RaR for progressive refinement and rediscovery of key structures. These modules interact in an end-to-end learning paradigm: PGI extracts interpretable and adaptive parts, CSA aligns them semantically, and RaR reinforces their discriminative strength, together forming a unified framework that bridges theoretical interpretability and practical classification performance.

C. Partial-Level Game-theoretic Interpretable (PGI) Module

Based on (4), a target instance is decomposed into multiple partial representations. By integrating (7), the partial-level interaction between two instances can be formulated as the cooperative game defined in (9), where $U_1 = \Psi^m_{\text{partial}}$ and $U_2 = \Psi^n_{\text{partial}}$

$$\mathcal{G}_1\left(\Psi^m_{\text{partial}}, \Psi^n_{\text{partial}}\right) \overset{\text{def}}{=} \sum_{i=1}^{\left|\Psi^m_{\text{partial}}\right|} \mathcal{S}\left(e_i; \Psi^m_{\text{partial}}\right) - \sum_{j=1}^{\left|\Psi^n_{\text{partial}}\right|} \mathcal{S}\left(e_j; \Psi^n_{\text{partial}}\right). \tag{9}$$

Equation (9) establishes a cooperative attribution view in which each partial is regarded as a participant contributing to the overall decision. In theory, Shapley-based contribution requires enumerating all possible coalitions, which is computationally intractable for dense visual features. Therefore, instead of exact enumeration, PGI implements a learnable surrogate mechanism that approximates marginal contribution through structured interaction and iterative aggregation.

Given an input optical remote sensing image x, the backbone extracts a feature map $X \in \mathbb{R}^{C\times H\times W}$. The feature map is reshaped into $X_{\text{flattened}} \in \mathbb{R}^{D\times N}$, where $N = H \times W$ and D denotes the reduced channel dimension. To preserve spatial correspondence, positional encoding is incorporated

$$X_{\text{pe}} = X_{\text{flattened}} + Position. \tag{10}$$

A set of learnable partial tokens P^h is initialized, where h denotes the number of partials. Each partial represents a potential contributor in the cooperative game. Their responses are updated via structured interaction with pixel-level features

$$l_Q\left(P^h\right) \in \mathbb{R}^{h\times D}, \quad l_K\left(X_{\text{pe}}\right) \in \mathbb{R}^{D\times N} \tag{11}$$

$$\mathcal{A} = \left(\frac{l_Q\left(P^h\right)\cdot l_K\left(X_{\text{pe}}\right)}{\sqrt{D}}\right) \tag{12}$$

$$\mathcal{A}' = \text{Softmax}\left(\mathcal{A}\right) \in (0,1)^{h\times N} \tag{13}$$

$$P^h{}_{\text{new}} = \text{LSTM}\left(\mathcal{A}' \cdot X^T{}_{\text{flattened}}, P^h{}_{\text{prev}}\right). \tag{14}$$

This update rule serves as a tractable surrogate for coalition-based contribution estimation. Instead of explicitly evaluating all feature subsets, the model learns contribution-aware aggregation through parameterized interactions. Each partial dynamically adjusts its representation according to its collaborative utility under the downstream matching objective, thereby approximating marginal contribution in a cooperative manner.

To further enforce contribution consistency, we introduce aggregation and separation constraints over the attention responses

$$\mathcal{L}_{\text{ag}} = \frac{1}{N}\sum_{i=1}^{h}\sum_{n=1}^{N}\left(\mathcal{A}^i_n - \overline{\mathcal{A}}^i\right)^2 \tag{15}$$

$$\mathcal{L}_{\text{se}} = \sum_{i\neq j}\frac{\mathcal{A}^i\cdot\mathcal{A}^j}{\|\mathcal{A}^i\|_2\|\mathcal{A}^j\|_2} \tag{16}$$

$$\mathcal{L}_{\text{attn}} = \alpha\mathcal{L}_{\text{ag}} + (1-\alpha)\,\mathcal{L}_{\text{se}} + \frac{1}{h\cdot N}\sum_{i=1}^{h}\sum_{n=1}^{N}\text{ReLU}\left(\mathcal{A}^i_n\right). \tag{17}$$

The aggregation term encourages internal coherence within each partial, while the separation term prevents redundant coalition formation across partials. Under this constrained interaction mechanism, the learned partials evolve into structured contributors whose weights reflect their relative importance in cooperative evidence construction. Consequently, PGI decomposes decision evidence into partial-level contributors, embedding interpretability into the model architecture rather than relying on posthoc explanation. The detailed pipeline of the Partial-Level Gametheoretic Interpretable (PGI) module is illustrated in Fig. 4.

D. CSA Module

Objects within the same subcategory may manifest in various forms across different samples, such as through differences in posture, scale, and appearance, leading to variations in their physical form. Despite these differences, the underlying semantic information remains constant. In few-shot fine-grained classification, the network needs to maintain invariance to any pose, scale, and appearance of the same subclass. Thus, we posit that objects of the same category share a common semantic. Drawing inspiration from the design philosophy of CLIP, we have developed a CSA Module that employs feature normalization, similarity matrix calculation, and adaptive alignment to align support and query image pairs of the same subcategory.

We have designed an innovative partial-level similarity matrix to calculate the similarity between query features and support features across the partial-level attention space. The process begins with reshaping the position-encoded features X_{pe} in order to obtain query features $F^Q \in \mathbb{R}^{D\times H\times W}$ and support features $F^S \in \mathbb{R}^{D\times H\times W}$. Then employing the

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partial-level attention $\mathcal{A}_{\text{reshape}} \in \mathbb{R}^{h\times H\times W}$ to weight the query features and support features

$$\tilde{F^Q} = \sum_{i=1}^{h} \mathcal{A}_i \odot F^Q, \quad \tilde{F^S} = \sum_{i=1}^{h} \mathcal{A}_i \odot F^S. \tag{18}$$

To quantify the degree of similarity between query features and support features in the attention space, we first compute the partial-level similarity matrix between the weighted features, followed by normalization to convert it into a probability distribution:

$$S_{\text{partial}} = \sigma\left(\frac{\tilde{F^Q}\cdot \tilde{F^S}^T}{\|\tilde{F^Q}\|\cdot\|\tilde{F^S}\|}\right). \tag{19}$$

In the equation above, the symbol σ denotes the softmax function. This similarity matrix is used to align the support features based on the query features. The framework of the CSA Module is shown in Fig. 5

$$F^{\text{aligned}} = S_{\text{partial}} \times F^S. \tag{20}$$

E. Refine-and-Rediscover (RaR) Module

Due to the small interclass differences and large intraclass differences in fine-grained data, the foreground target of different subcategories are similar, while the background of same subcategory may be different. According to (5) and (8), we found that there is a game between the background and the target partials of remote sensing images. This battle between the most critical target partials and useless information in background determines the decision-making process for few-shot fine-grained classification. The game can be defined as shown in the following equation:

$$\begin{aligned}\mathcal{G}_2\left(\Psi^m_{\text{partial}}, \Psi_{\text{background}}\right) = &\sum_{i=1}^{|\Psi|} \mathcal{S}\left(e_i; \Psi\right) \\ &- \sum_{r=1}^{t-1}\sum_{i=1}^{|\Psi^{n_r}_{\text{partial}}|} \mathcal{S}\left(e_i; \Psi^{n_r}_{\text{partial}}, n_r \neq m\right) \\ &- \sum_{j=1}^{|\Psi^m_{\text{partial}}|} \mathcal{S}\left(e_j; \Psi^m_{\text{partial}}\right).\end{aligned} \tag{21}$$

The game between background and partials proves that redundant information in the background can interfere with the learning of network for fine-grained features. In order to ensure the focus on partials of subject and making decisions through partials matching, it is necessary to decouple the discriminative subject from the background. Based on the partial-level attention features obtained from PGI, we propose a RaR Module for remote sensing images to remove redundant and interference factors in background, thus encourage the network to capture the most critical subtle features (that is, essential features) for fine-grained classification.

A progressive interactive training strategy is used to explore the region of fine-grained objects in remote sensing images. Set two adjustable threshold parameters, t_c and t_d. According to the position-encoded partial-level attention features extracted in PGI, RaR using the weight of each element in feature map and threshold parameter t_c to select the main subject, fill the background with mask and resize the original input to obtain cropped image x_{crop}. These cropped samples make the model more focused on the subtle differences in the main area of objects.

First reshape the partial-level attention features to recover its position information and obtain overall attention from h partials

$$\overline{\mathcal{A}_{\text{reshape}}} = \frac{1}{h}\sum_{i=1}^{h} \mathcal{A}_i. \tag{22}$$

Then we compute a per-image percentile threshold that automatically adapts to the dynamic value range of given activation map

$$\tau\left(\mathcal{A}; t\right) = \text{Percentile}_{t\times 100}\left(\{\mathcal{A}\left(u,v\right)\}_{u,v}\right). \tag{23}$$

Instead of employing a fixed numerical threshold, the threshold $\tau(\mathcal{A}; t)$ is defined as the tth empirical percentile of all pixel responses in $\mathcal{A}$. Next we traverse the coordinates of $\overline{\mathcal{A}_{\text{reshape}}} \in \mathbb{R}^{H\times W}$ to generate binary masks for cropping and dropping based on the subject selection thresholds

$$M_c\left(i,j\right) = \begin{cases} 1, & \text{if } \overline{\mathcal{A}_{\text{reshape}}}\left(i,j\right) > \tau\left(\overline{\mathcal{A}_{\text{reshape}}}; t_c\right) \\ 0, & \text{otherwise.}\end{cases} \tag{24}$$

Among them, i and j represent the horizontal and vertical coordinates of elements in the feature map, respectively.

Besides, define an operation function $\mathbb{I}$ to extract each connected region of mask M_c, select the area with the most pixels. The selected region is scaled according to the scaling parameter s. Subsequently, this scaled region is multiplied element-wise with the original input image to obtain the cropped sample

$$x_{\text{crop}} = \left(s \times \text{argmax}_{\mathbb{I}(M_c)_{\max}} \sum_{\text{pixel}} \mathbb{I}\left(M_c\right)\right) \odot x. \tag{25}$$

At the same time, another mask based on threshold t_d is generated. This mask delineates the regions of high salience within the original samples, thereby directing the network's attention to alternative feature domains within the image. By doing so, it encourages the network learn more comprehensive feature representations. This approach mitigates the model's reliance on specific background elements or potential sources of interference, thereby enhancing its capacity for generalization. The overall process of the Refine-and-Rediscover (RaR) module is presented in Fig. 6

$$M_d\left(i,j\right) = \begin{cases} 0, & \text{if } \overline{\mathcal{A}_{\text{reshape}}}\left(i,j\right) > \tau\left(\overline{\mathcal{A}_{\text{reshape}}}; t_d\right) \\ 1, & \text{otherwise}\end{cases} \tag{26}$$

$$x_{\text{drop}} = s \times M_d \odot x. \tag{27}$$

F. Training Strategy

In this work, we define a N-way, K-shot few-shot fine-grained classification task for remote sensing images. The dataset is partitioned into a base set $\mathcal{D}_{\text{base}}$ and a novel set $\mathcal{D}^{\text{novel}}$, where $\mathcal{D}^{\text{base}} \cap \mathcal{D}^{\text{novel}} = \emptyset$. The base set is utilized during training, while the novel set is employed during testing. Samples are drawn from these sets to create support-query

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pairs, $\mathcal{D}^S = \{(x_{i,j}, y_{i,j}) \mid i = 1, \ldots, N; j = 1, \ldots, K\}$ and $\mathcal{D}^Q = \{(x_j, y_j) \mid j = 1, \ldots, |\mathcal{D}^Q|\}$. Among them, x is the remote sensing image, y is the sample category label. During the training process, the support-query pairs are reformed as crop and drop samples through the aforementioned strategies. Subsequently, a progressive training strategy is employed, which is implemented through iterative constraints on the loss.

In the training phase (and only in the training phase), features are learned that are relevant to the entire base category space, thereby enriching interclass relationships. Specifically, a linear fully connected layer is appended to the aligned query features, and the output of this layer is subject to a cross-entropy loss constraint with the true labels, resulting in a global loss. This approach facilitates the optimization of the model by leveraging the discriminative information encapsulated within the base categories

$$\mathcal{L}_{\text{glo}} = \beta \cdot \text{CrossEntropyLoss}\left(\text{FC}\left(x_{\text{drop}}\right), y\right) + (1-\beta) \cdot \text{CrossEntropyLoss}\left(\text{FC}\left(x_{\text{crop}}\right), y\right). \quad (28)$$

The hyperparameter β is designed to modulate the influence of the classification outcomes derived from dropped and cropped samples on the global loss, denoted as $\mathcal{L}_{\text{glo}}$. Thus, for any support-query sample pair in training set, we can calculate the similarity between query features and support features in the spatial dimension, and assigning queries to the most similar support category

$$f\left(x_{i,j}^S, x^Q\right) = \arg\max_i \left(\frac{1}{k}\sum_{j=1}^{K} \cos\left(F^Q, F_{i,j}^{\text{aligned}}\right)\right). \quad (29)$$

The similarity between query and support samples is computed using both the original and cropped images, these are then compared with the ground truth labels to compute the loss term $\mathcal{L}_{\text{similarity}}$, which serves as a constrain in cross-pose alignment module, transforming support and query subjects with different postures into partial-level aligned features

$$\mathcal{L}_{\text{similarity}} = \gamma \cdot \text{CrossEntropyLoss}\left(f\left(x_{\text{crop}}^S, x_{\text{crop}}^Q\right), y^Q\right) + (1-\gamma) \cdot \text{CrossEntropyLoss}\left(f\left(x^S, x^Q\right), y^Q\right) \quad (30)$$

where γ is weighting factors. Above all, the overall loss is formulated as follows:

$$\mathcal{L}_{\text{overall}} = \lambda_1 \cdot \mathcal{L}_{\text{attn}} + \lambda_2 \cdot \mathcal{L}_{\text{glo}} + \lambda_3 \cdot \mathcal{L}_{\text{similarity}}. \quad (31)$$

IV. Experiment

In this section, we delineate the datasets utilized, the configuration of experimental hyperparameters. Conducting comparative analyses with baseline methods, outcomes across varying hyperparameters, and further discussion.

A. Datasets

To comprehensively evaluate the effectiveness and generalization capability of the proposed few-shot fine-grained classification model, experiments were conducted on five datasets featuring diverse characteristics, encompassing both natural and remote sensing imagery. Among them, FGVC-Aircraft [47], FGSC-23 [45], and FGSCR-42 [46] are fine-grained remote sensing datasets focusing on aircraft and ship classification, respectively. CUB-200-2011 [48] serves as a representative natural image dataset dedicated to fine-grained bird classification, demonstrating the model's ability to extend from remote sensing fine-grained classification to natural image domains. Collectively, these datasets verify the model's capacity to capture discriminative details across different remote sensing targets, and its transferability from remote sensing to natural imagery. The detailed statistics of these datasets are summarized in Table I. All datasets were reorganized by repartitioning categories into mutually exclusive training, validation, and testing sets to ensure nonoverlapping classes. This reorganization enables rigorous evaluation under few-shot settings, where the model must generalize to previously unseen variants based on only a limited number of annotated examples.

1) *FGVC-Aircraft [47]:* The FGVC-Aircraft dataset originally contains 10 000 high-quality images of aircraft. To adapt this dataset for few-shot classification, each image was first assigned to its corresponding variant category using the official CSV annotations. The 100 variant classes were then reorganized into mutually exclusive training, validation, and test subsets, ensuring that categories in the validation and test sets are entirely novel and never seen during training.
2) *FGSC-23 [45]:* The FGSC-23 dataset, a high-fidelity optical remote sensing resource for fine-grained ship classification, comprises 23 distinct ship categories. The FGSC-23 offers a rich tapestry of diverse maritime scenes and is distinguished by its meticulous categorization and comprehensive labeling, setting it apart from other optical remote sensing datasets focused on ship target classification. Illustrative examples of the 23 ship classes within the FGSC-23 are presented in Fig. 7.
3) *FGSCR-42 [46]:* The FGSCR-42 dataset, an extensive compilation for fine-grained ship classification in remote sensing [46], encompasses a detailed taxonomy of 42 categories. Constructed from detection datasets such as DOTA [49] and HRSC2016 [50], its cropped image patches emphasize detailed interclass differences among ship types, offers a more granular classification compared to FGSC-23. Representative ship image slices from the FGSCR-42 dataset are depicted in Fig. 8.
4) *CUB-200-2011 and CUB-200-2011-bb [48]:* To assess the adaptability of the proposed framework to natural image domains, we further conducted experiments on the CUB-200-2011. The CUB-200-2011 dataset comprises 11 788 bird images from 200 species. It serves as a well-established benchmark for fine-grained visual categorization. We also use its bounding-box cropped images that is the CUB-200-2011-bb dataset. We use the same class split as FRN [23], which utilizes 100, 50, and 50 classes for training, validation, and testing.

B. Implementation Details

We employed a widely recognized backbone network, ResNet-18, to feed preprocessed images into the main trunk

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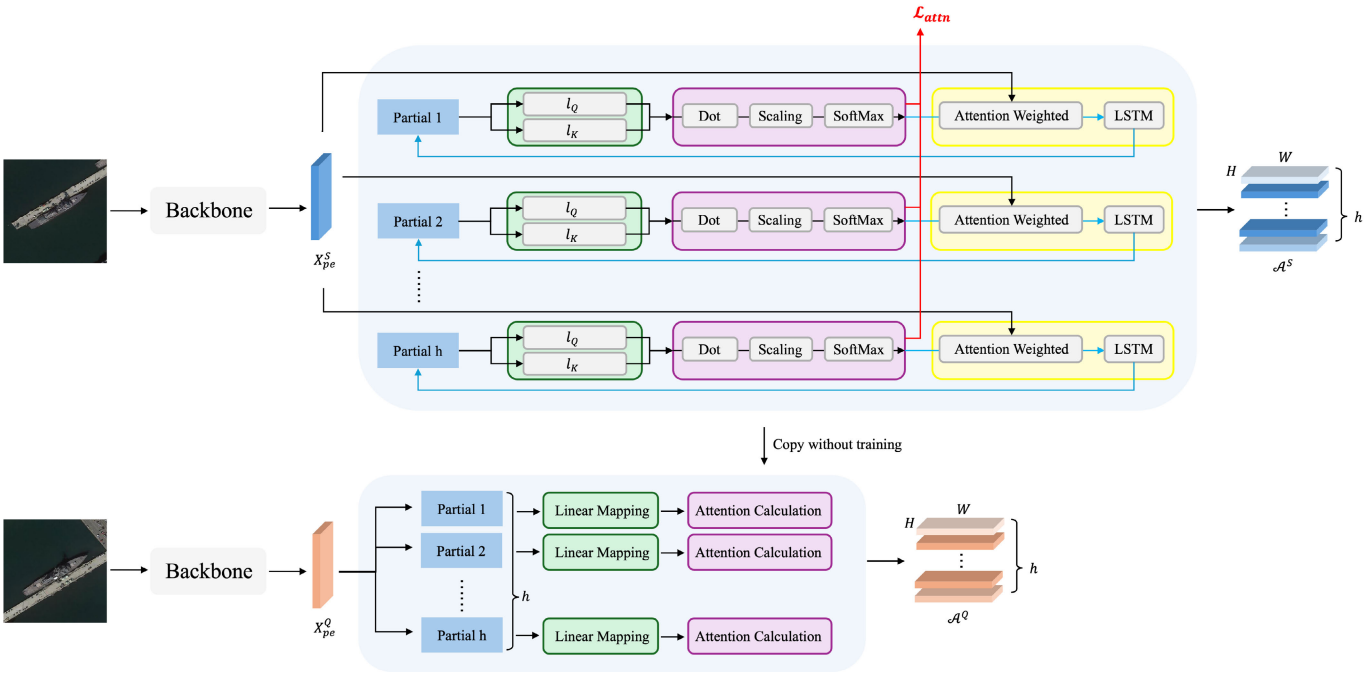


Fig. 4. Flowchart of the PGI module. Through the attention mechanism, the partial heads, set to a parameter denoted as h, are updated using LSTM, enabling the module to dynamically learn fine-grained features. Blue arrows indicate processes used exclusively during training, while red arrows denote components of the loss function.

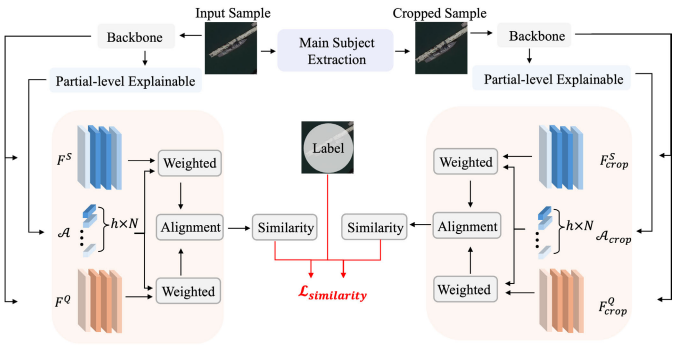


Fig. 5. Cross-pose Semantic Alignment (CSA) Module enhances the similarity between support and query samples by weighting their features through the intermediate variables output by the PGI module. In our experiments, we applied the CSA module to both original and cropped samples obtained via the RaR module. By calculating the similarity between aligned support-query image pairs and integrating these similarities, we support the final classification decision.

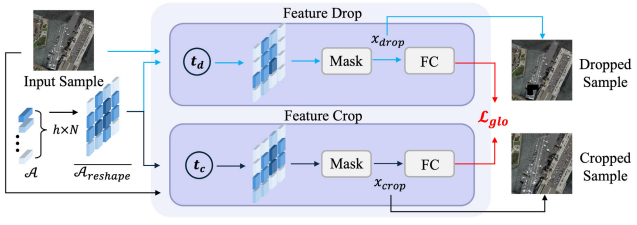


Fig. 6. RaR process. The two pathways generate cropped samples and dropped samples, respectively. Blue arrows indicate processes used exclusively during training, while red arrows denote components of the loss function.

for feature extraction. The model consists of two stages: the original stage and the refinement and rediscovery stage. Both stages share the parameters of the feature extractor, while the global classifier maintains independent parameters. In the original stage, the raw image is fed into the backbone network and the PGI Module to extract feature representations F. In the refinement stage, the input is the refined image generated by the RaR Module, from which refined features F' are obtained. The feature representations from both stages are aligned through the CSA Module to ensure semantic consistency. Finally, the similarity scores computed in the two stages are integrated to form the final decision.

In the experimental setup, several hyperparameters were carefully tuned to ensure optimal performance. The input images were first preprocessed through resizing and normalization before being fed into the model. The initial learning rate was set to 0.001, with a step size of 10 and a weight decay of 0.0001. The AdaBelief optimizer was adopted for stable convergence, and the batch size was fixed at 64. Regarding the model configuration, we introduced an adjustable hyperparameter h, termed Partial Number (**PN**), which determines the number of learnable parts. This parameter plays a crucial role in guiding the attention mechanism to capture discriminative fine-grained regions. Based on experimental analysis, h was set to 5. To further evaluate the effectiveness of the proposed RaR Module, we conducted a series of ablation studies to examine the influence of the threshold parameters t_c and t_d. These two parameters, respectively, control the degree of region preservation in the Refine stage and the masking intensity in the Rediscover stage. The goal of the Refine process is to suppress redundant background while emphasizing the most informative object regions. After extensive experimentation, t_c was set to 0.7 while t_d was set to 0.8. Moreover, through comprehensive experiments, we observed

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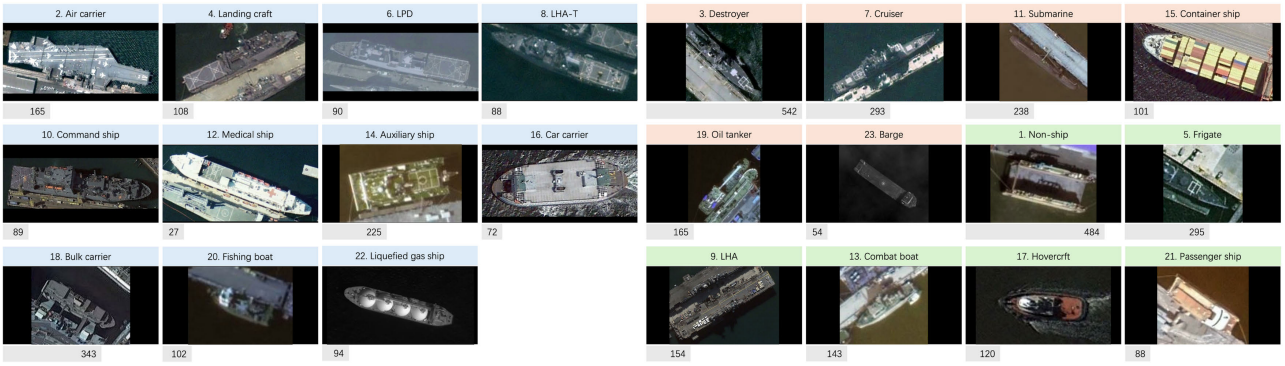


Fig. 7. FGSC-23 dataset contains samples from 23 fine-grained ship classes [45]. To adapt the dataset for few-shot learning, we reconstructed it by splitting the original categories randomly into base and novel classes in a 3:1 ratio. The bar above the legend displays the true category of the sample and represents the division of the small sample dataset with colors. Specifically, blue represents the training set, orange represents the validation set, and green represents the test set. The training set and test set together form the base categories in small sample training, while the test set is the novel categories. Below each instance sample, the legend shows the proportion of the category in the dataset with a gray bar and indicates the precise number of samples. Abbreviations include LPD for amphibious transport dock, LHA-T for Tarawa-class amphibious assault ship, and LHA for amphibious assault ship. This dataset presents several highly similar subcategories (e.g., #5 and #7, #6 and #8) and substantial color variation within instances of the same class. Additionally, some targets, such as #8 and #22, are heavily affected by scene interference.



Fig. 8. FGSCR-42 dataset comprises samples from 42 fine-grained ship classes [46]. Following the same protocol as with FGSC-23, we adapted FGSCR-42 for few-shot learning by randomly splitting the original dataset classes into base and novel categories in a 3:1 ratio. The bar above the legend displays the true category of the sample and represents the division of the small sample dataset with colors. Specifically, blue represents the training set, orange represents the validation set, and green represents the test set. The training set and test set together form the base categories in small sample training, while the test set is the novel categories. Below each instance sample, the legend shows the proportion of the category in the dataset with a gray bar and indicates the precise number of samples. This dataset features a wide range of categories, diverse samples, and complex scenes, making it a valuable resource for fine-grained classification tasks in remote sensing imagery.

that the magnitude of the loss function defined in (31) of Section III should remain consistent across training. Consequently, the hyperparameters λ_1, λ_2, and λ_3 were all configured to 1. The detailed process of hyperparameter tuning and the corresponding results can be found in Section IV-F of this chapter.

The experimental setup was conducted on an Ubuntu 16.04 operating system, equipped with dual NVIDIA RTX 3090 GPUs to facilitate high-performance computing. The model architecture was developed utilizing the PyTorch framework, leveraging its robust capabilities for deep learning research and development.

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TABLE I
STATISTICS OF THE DATASETS USED IN OUR EXPERIMENTS

	FGVC-Aircraft [47]		FGSC-23 [45]		FGSCR-42 [46]		CUB-200 [48]		CUB-200-bb	
Subset	**Categories**	**Images**	**Categories**	**Images**	**Categories**	**Images**	**Categories**	**Images**	**Categories**	**Images**
Train	50	5000	11	1404	21	4687	100	5903	100	5903
Val	25	2500	6	1392	11	2318	50	2927	50	2927
Test	25	2500	6	1285	10	2315	50	2958	50	2958
Total	100	10000	23	4081	42	9320	200	11788	200	11788

TABLE II
COMPARISON WITH FEW-SHOT FINE-GRAINED SOTA METHODS ON REMOTE SENSING IMAGES (5-WAY CLASSIFICATION)

Method	**Venue**	**Accuracy**									
		FGVC-Aircraft [47]		FGSC-23 [45]		FGSCR-42 [46]		CUB-200 [48]		CUB-200-bb	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
FRN [23]	CVPR'21	46.19	62.55	34.13	40.08	45.47	71.20	63.84	78.13	69.45	85.16
TDM [8]	CVPR'22	41.74	61.70	33.64	39.72	43.74	62.07	71.37	86.45	74.39	88.89
FA-FRN [51]	TGRS'22	46.44	63.00	-	-	46.76	74.58	65.91	80.87	-	-
CW-FRN [52]	TGRS'23	47.43	65.98	-	-	49.97	74.46	65.41	80.48	-	-
RI-FRN [10]	TGRS'24	51.86	68.16	-	-	53.74	79.66	67.97	83.44	-	-
MLI [53]	TIP'24	-	-	40.92	46.79	55.69	77.88	**75.29**	86.49	**78.20**	89.45
Ours	-	**57.79**	**70.81**	**41.28**	**48.94**	**57.83**	**79.89**	72.16	**86.74**	74.60	**89.94**

C. Comparisons With SOTA Methods

To comprehensively evaluate the proposed method, experiments were conducted on five benchmark datasets, including three remote sensing datasets (FGVC-Aircraft, FGSC-23, and FGSCR-42) and two natural image datasets (CUB-200 and CUB-200-bb). These reorganized few-shot datasets cover fine-grained classification of remote sensing images (specifically aircraft and ship targets), as well as natural images. The evaluation follows the standard five-way one-shot and five-shot protocols. Comparative experiments were performed against state-of-the-art (SOTA) few-shot fine-grained classification methods (specifically bird targets), where the upper part of Table II presents remote sensing-oriented methods (FRN [23], TDM [8], FA-FRN [51], CW-FRN [52], RI-FRN [10]) and the lower part shows a representative method primarily designed for natural image classification (MLI [53]).

1) Remote Sensing Fine-Grained Classification: As shown in Table II, our method achieves the best performance on all three remote sensing fine-grained datasets, establishing new SOTA results. Specifically, the proposed approach attains accuracies of 57.79%/70.81%, 41.28%/48.94%, and 57.83%/79.89% on FGVC-Aircraft, FGSC-23, and FGSCR-42 under the 1-shot and 5-shot settings, respectively. These results demonstrate that our model effectively captures discriminative semantic components across fine-grained target types. Benefiting from the PGI module, which learns interpretable partial representations through attention-driven decomposition, and the CSA module, which aligns discriminative features across varying poses and scales, the model achieves robust and semantically consistent matching. Furthermore, the RaR module enhances feature refinement by suppressing background noise and rediscovering informative cues, resulting in improved generalization and discriminative capability.

2) Natural Image Fine-Grained Classification: When extended to natural image datasets, our method achieves superior generalization. On CUB-200 and CUB-200-bb, the proposed approach attains accuracies of 72.16%/86.74% and 74.60%/89.94%, respectively, surpassing all remote sensing-oriented methods. Specifically, our model outperforms the best remote sensing baseline (RI-FRN 67.97%/83.44%) by 4.19% and 3.3% on CUB-200 under the 1-shot and 5-shot settings, respectively. Moreover, we extended the experiments to natural image few-shot fine-grained classification tasks using the CUB-200 and CUB-200-bb datasets. As shown in Table III, our method remains highly competitive compared with the latest fine-grained classification approaches, including Bi-FRN [21], MLI [53], and BFCCN [58]. Specifically, IPM-Net achieves accuracies of 72.16%/86.74% on CUB-200 and 74.60%/89.94% on CUB-200-bb under the 1-shot and 5-shot settings, respectively. Compared with Bi-FRN [21], which attains 74.36%/88.64% on CUB-200 and 75.86%/89.89% on CUB-200-bb, IPM-Net achieves comparable or slightly higher performance while using a unified framework originally designed for remote sensing imagery. Furthermore, the model narrows the performance gap with the current best-performing

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TABLE III
COMPARISON WITH FEW-SHOT FINE-GRAINED SOTA METHODS ON NATURAL IMAGES (5-WAY CLASSIFICATION)

Method	Venue	Accuracy			
		CUB-200 [48]		CUB-200-bb	
		1-shot	5-shot	1-shot	5-shot
FRN [23]	CVPR'21	63.84	78.13	69.45	85.16
TDM [8]	CVPR'22	71.37	86.45	74.39	88.89
FA-FRN [51]	TGRS'22	65.91	80.87	-	-
CW-FRN [52]	TGRS'23	65.41	80.48	-	-
RI-FRN [10]	TGRS'24	67.97	83.44	-	-
GLFA [54]	PR'23	76.52	**90.27**	-	-
SAPENet [55]	PR'23	70.38	84.47	-	-
Bi-FRN [21]	AAAI'23	74.36	88.64	75.86	89.89
MLI [53]	TIP'24	75.29	86.49	78.20	89.45
BAML [56]	ESWA'24	-	-	68.55	84.77
AttrDD [57]	TIP'25	54.52	-	-	-
BFCCN [58]	PRLetters'25	**77.13**	89.22	**80.93**	**90.94**
Ours	-	72.16	86.74	74.60	89.94

method BFCCN [58], with only a 0.29% difference in 5-shot accuracy on CUB-200-bb (89.94% versus 90.94%).

These results demonstrate that IPM-Net, though developed for few-shot fine-grained classification in remote sensing, also exhibits strong transferability to natural image domains. It is the interpretable architecture enables the network to capture semantically consistent and discriminative regions even in complex natural scenes. By adding partial-level decomposition and semantic consistency during training, the model learns disentangled, localized features that capture object-centric cues rather than dataset-specific texture artifacts; this representation is naturally transferable to natural images where discriminative partials dominate. Moreover, the training-time refinement and rediscovery mechanisms encourage the backbone to discover complementary cues and to be resilient to occlusion or background noise, which improves generalization across both remote sensing and natural scenes.

Besides, in the remote sensing scene, where interclass differences are subtle and training samples are scarce, such partial-wise, interpretable representations act as a strong regularizer, who reduce overfitting, concentrate learning capacity on stable, discriminative subregions, and make few-shot matching more reliable—hence the observed SOTA gains.

D. Interpretability Analysis

IPM-Net, designed with inherent Interpretability through the PGI module, offers transparency into how the model learns and transfers knowledge in few-shot fine-grained classification tasks. The PGI module allows us to gain insight of the features learned from support samples and how these features are applied to classify query samples, providing valuable insight into the network's internal workings. Use the concept of partial level decision making to qualitatively explain IPM-Net. According to the formulation in (4), PGI represents each target as a combination of multiple learnable particles, these partials are used to encode and align features from both the support and query sets, enabling the network to transfer knowledge during few-shot classification. We examine the interpretability of our method through both qualitative and quantitative analyses.

1) Qualitative Analyses: To quantitatively evaluate the performance of interpretability analysis methods, we use the following evaluation metrics: Average Drop, Coherency, Complexity, and Average DCC (ADCC) [59]. The variable y_c represents the output score for class c when the full image is utilized. In contrast, o_c denotes the output score for the same class c when an explanation map is applied. The function E_c signifies the saliency map produced by an interpretability analysis method or module, and x stands for the input image.

1) Average drop measures the average percentage decrease in confidence for the target class c when the model only sees the explanation map rather than the full image

$$\text{AverageDrop}(x) = \left(\frac{\max(0, y_c(x) - o_c(x))}{y_c(x)}\right) \times 100\%. \tag{32}$$

2) Maximum coherency evaluates whether the saliency map of an interpretability analysis method contains all relevant features and removes irrelevant ones in a coherent manner when explaining the prediction. Therefore, for a given input image and the class of interest, the saliency map should not change when conditioned on the saliency map itself

$$E_c(x \odot E_c(x)) = E_c(x) \tag{33}$$

$$\text{Coherency}(x) = \frac{\text{Cov}(E_c(x \odot E_c(x)), E_c(x))}{\sigma_{E_c(x \odot E_c(x))}\sigma_{E_c(x)}}. \tag{34}$$

3) Minimum complexity requires the saliency map to be as simple as possible, meaning it should include the fewest pixels necessary to explain the prediction. The complexity is minimized when the attribution method highlights only a small number of pixels. This metric helps assess the conciseness of the interpretability method, i.e., using the fewest pixels to explain the prediction

$$\text{Complexity}(x) = \|E_c(x)\|_1. \tag{35}$$

4) ADCC is used for interpretability evaluation. ADCC combines multiple aspects, including average drop, coherency, and complexity, to provide a comprehensive and fair assessment of the interpretability of a method

$$ADCC(x) = 3\left(\frac{1}{\text{Coherency}(x)} + \frac{1}{1 - \text{Complexity}(x)} + \frac{1}{1 - \text{AverageDrop}(x)}\right)^{-1}. \tag{36}$$

To quantitatively assess the interpretability of the proposed PGI module, experiments were conducted under the *5-way 1-shot* setting on the FGSC-23 dataset. Two representative posthoc interpretability analysis methods, Grad-CAM and Score-CAM, were selected as baselines. Grad-CAM is one of the most widely adopted visualization techniques that highlights salient regions contributing to a model's decision through class-specific gradient information. Score-CAM, as

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TABLE IV
QUANTITATIVE INTERPRETABILITY COMPARISON OF PGI AND OTHER METHODS ON TWO DATASETS

Dataset	Method	Avg Drop↓	Coherence↑	Complexity↓	ADCC↑
FGSC-23 [45]	Grad-CAM	71.25	59.37	25.68	46.11
	Score-CAM	58.74	63.09	49.23	50.16
	PGI	**36.24**	**92.68**	**18.34**	**77.49**
FGSCR-42 [46]	Grad-CAM	63.82	65.57	24.79	58.99
	Score-CAM	44.79	71.30	38.54	62.66
	PGI	**29.01**	**94.13**	**17.41**	**82.57**

an improved variant, eliminates the dependence on gradients by leveraging forward activations to enhance stability and localization accuracy.

As shown in Table IV, the proposed PGI module consistently achieves the best performance across all metrics on both datasets. For FGSC-23, PGI attains an Average Drop of 36.24, a Coherency of 92.68, a Complexity of 18.34, and an ADCC of 77.49, substantially outperforming Grad-CAM (46.11 ADCC) and Score-CAM (50.16 ADCC). For FGSCR-42, PGI further demonstrates superior interpretability, achieving an Average Drop of 29.01, a Coherency of 94.13, a Complexity of 17.41, and an ADCC of 82.57, again surpassing Grad-CAM (58.99 ADCC) and Score-CAM (62.66 ADCC) by a clear margin. These results indicate that the PGI module provides more coherent and concise visual explanations while minimizing redundancy and emphasizing essential discriminative regions. This superior performance can be attributed to the intrinsic design of the PGI module, which is embedded within the IPM-Net architecture rather than applied post hoc. The PGI module establishes a direct correspondence between feature activation and decision-making during training, allowing the model to learn interpretable and semantically aligned features in an end-to-end manner. In contrast, Grad-CAM and Score-CAM are external visualization tools that rely on backpropagation or activation reweighting after model convergence, often introducing noise or missing subtle fine-grained cues critical in few-shot scenarios.

Overall, the consistent advantage of the PGI module across all interpretability metrics demonstrates that the proposed IPM-Net possesses strong intrinsic interpretability. The PGI module not only improves the transparency of the model's decision process but also reinforces the alignment between semantic reasoning and visual evidence, yielding interpretable and reliable explanations suitable for few-shot fine-grained classification tasks.

2) Quantitative Analyses: By visualizing the feature $\mathcal{A}$, as defined in (12), we were able to gain valuable insights into how the model focuses on different fine-grained components of the objects, effectively capturing and transferring discriminative knowledge from support to query.

Fig. 9 presents visualizations of support and query images from a 5-way 1-shot task on the FGSCR-42 dataset. The second column displays the aggregated overall attention, denoted as $\mathcal{A}'$, while the subsequent columns depict activations corresponding to individual partials. In Fig. 9, we present an example involving a fuzzy query sample, with partial visualizations highlighting the fine-grained features learned by the network in recognizing the Akizuki-class destroyer. It shows that in the overall attention map, the network effectively captures prominent ship features in the clear support samples, whereas in the fuzzy query samples, the attention shifts predominantly to the stern, with relatively weaker focus on the bow. This is likely because the stern in the fuzzy sample retains more distinguishable features. Specifically, the activations corresponding to Partial 1 are primarily concentrated at the stern, where twin-mounted main guns (rear turrets) or depth charge delivery devices are located. Partial 4 highlights the main guns located at both the bow and stern, consisting of two twin-mounted 100 mm dual-purpose guns for anti-air and surface engagements. Finally, Partial 5 focuses on the bridge structure, which serves as a significant classification feature due to its tall and prominent shape. In addition to this, it can be observed that certain partials (e.g., partial 2 and partial 3) do not focus on the target object itself but rather on the surrounding area. These partials learn texture information of the target and capturing the hull, contour, size, or background, which is also crucial for the effective utilization of all available information in small-sample classification. We will discuss the specific circumstances in the section on hyperparameter tuning.

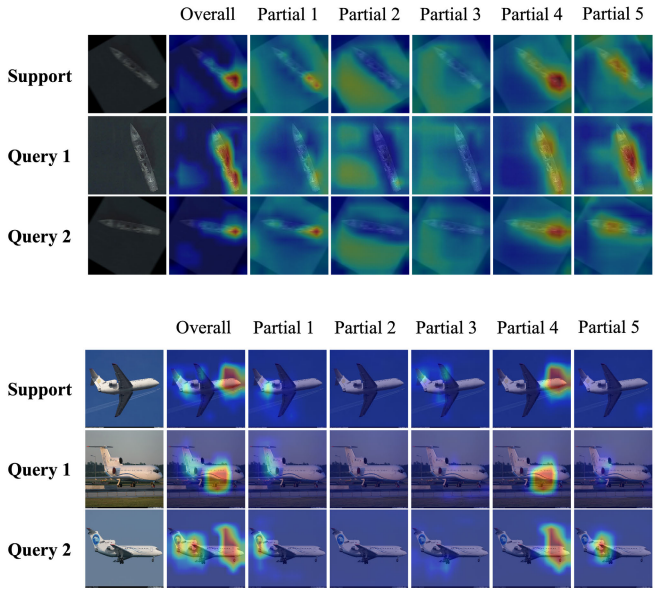


Fig. 9. Visualization of partial-level attention in IPM-Net for few-shot fine-grained classification on FGSCR-42 and FGVC-Aircraft. The first line is a sample from the support set, while the second line and the third line are the corresponding query images. The second column visualizes the overall attention, while the following columns highlight activations of the five partials, each representing specific ship or aircraft components (e.g., mast, deck, and bridge). These visualizations demonstrate how the model aligns support and query features for fine-grained classification.

These results provide clear evidence that IPM-Net effectively captures and transfers fine-grained structural features in remote sensing images during few-shot learning. The model's

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TABLE V
ABLATION STUDY FOR THE INTERPRETABILITY MECHANISM

Dataset	Accuracy (%)			
	Baseline	Standard	PGI	Improve
FGSC-23	37.21	40.97	**41.28**	0.31
FGSCR-42	46.07	55.54	**57.83**	2.29
FGVC-Aircraft	46.23	53.62	**57.79**	4.17

innovative partial-level framework enhances its ability to fine-grained classification by isolating and leveraging critical partial-level details, addressing the lack of interpretability in few-shot fine-grained classification networks for remote sensing image.

3) Effectiveness of the Interpretability Mechanism: To evaluate the validation of the interpretability mechanism and the actual contribution of the game-theoretic (Shapley value-based) modeling, we conducted an ablation study that compared three configurations to isolate the effects of different components. We explicitly contrast three configurations, and analyzing the impact on classification performance: 1) baseline without PGI, where backbone features feed directly into subsequent modules; 2) a variant using standard attention without Shapley-based contribution modeling; and 3) our full PGI module with cooperative game formulation. This approach eliminates the task-specific adaptive learning of partial features provided by game-theoretic (Shapley value-based) modeling, thereby allowing a direct comparison of the performance differences.

The results of the ablation experiments, shown in Table V, demonstrate that while standard attention provides modest gains, the Shapley value-based formulation that is PGI achieves substantially larger improvements. This performance gap between attention-only and Shapley-guided attention directly validates that the cooperative game framework contributes independently beyond simple attentional weighting. Specifically, the model with PGI's classification accuracy surpasses 2.29% further than the standard attention mechanism on the FGSCR-42 dataset and surpasses 4.17% on the FGVC-Aircraft dataset. These results highlight that PGI with its Shapley-based modeling mechanism plays a crucial role in exploring interpretable, task-specific partial-level essential features. The substantial increase in FGSCR-42 and FGVC-Aircraft suggest that PGI's adaptive approach to feature decomposition is particularly beneficial in handling more complex datasets with intricate structures.

Further analysis reveals the importance of PGI's dynamic partial exploring mechanism. In the absence of PGI, the model relies solely on fixed feature extraction and matching, which limits its ability to adapt to diverse object structures. For instance, objects with complex features (e.g., aircraft with detailed wing assemblies) require a flexible approach to break down the object into more informative and discriminative parts. Without PGI, the network's fixed feature representations may either over-segment simple regions or fail to capture critical structural details, as evidenced by the significant accuracy drop in the more complex dataset.

TABLE VI
COMPARATIVE PRECISION FOR MODULE ABLATION EXPERIMENTS UNDER THE 5-WAY-1-SHOT SETTING

Dataset	PGI	RaR	CSA	Accuracy(%)
FGSC-23	-	✓	✓	40.97
	✓	-	✓	39.43
	✓	✓	-	34.12
	✓	✓	✓	**41.28**
FGSCR-42	-	✓	✓	55.54
	✓	-	✓	53.24
	✓	✓	-	49.67
	✓	✓	✓	**57.83**

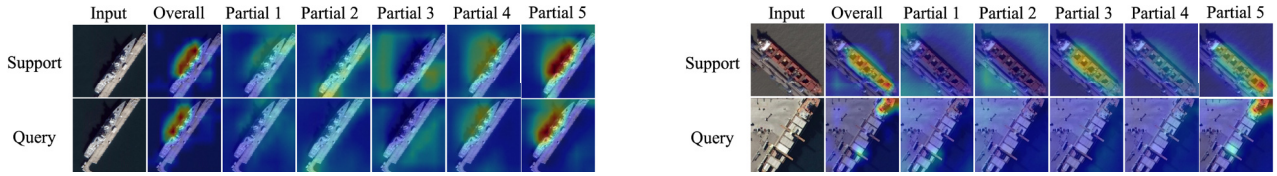
E. Ablation Study

To validate the effectiveness of each proposed module, we conducted ablation experiments as shown in Table VI.

1) RaR Module: We conducted an ablation study on the RaR module, a training-phase data refinement strategy specifically designed for remote sensing imagery. To evaluate its impact, we compared an IPM-Net model trained without cropping and rediscovery operations to one utilizing the RaR module. The absence of the RaR module led to accuracy decreases of 1.85% on the FGSC-23 dataset and 4.34% on the FGSCR-42 dataset, as shown in Table VI, highlighting the module's significance in mitigating background interference. Furthermore, we used the PGI module to analyze the interpretability of models trained with and without the RaR strategy. Visualization in Fig. 10(a) illustrates the effect of the RaR module using an example of an Akizuki-class destroyer. Without RaR during training, the learned partial regions became significantly larger and less refined, failing to capture fine-grained features effectively. These regions contained excessive background information, which likely interfered with classification. In contrast, models trained with RaR exhibited more precise and discriminative attention, focusing on critical structural details such as hull contours and deck layouts. This demonstrates that the RaR module enhances the network's capacity to emphasize relevant foreground regions, thereby improving interpretability and generalization in few-shot fine-grained classification tasks.

2) CSA Module: We removed the Cross-pose Semantic Alignment (CSA) Module and replaced the proposed partial-level similarity metric with a simpler cosine similarity for few-shot learning. Quantitative results reveal significant performance degradation, classification accuracy drops by 7.16% on the FGSC-23 dataset and 8.91% on the FGSCR-42 dataset. This pronounced decline highlights the indispensable role of CSA in resolving pose and viewpoint discrepancies when learning partial features. Without CSA, the RaR-based refinement fails to establish accurate semantic correspondences between support and query partials under varying poses,

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(a) Visualization of RaR module ablation experiment.

(b) Visualization of CSA module ablation experiment.

Fig. 10. Visualization of IPM-Net during ablation experiments on CSA and RaR modules, respectively, where (a) partial-level visualization of supply and query samples when the RaR module does not exist and (b) visualization results of the CSA module ablation experiment, respectively.

TABLE VII

ACCURACY (%) OF IPM-NET WITH VARYING PARTIAL NUM ON FGSC-23 AND FGSCR-42 DATASETS

PN	FGSC-23		FGSCR-42	
	5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
1	32.63	45.28	52.95	68.85
3	38.26	**48.37**	55.26	69.20
5	**41.28**	47.94	**57.83**	**70.39**
10	40.12	47.03	54.39	68.23

leading to misaligned feature representations that obscure subtle discriminative patterns. Qualitative visualizations further demonstrate that the absence of CSA results in incoherent attention maps, where critical fine-grained regions are either overlooked or mismatched across poses. Visualizations in Fig. 10(b) illustrates a container ship example where, in the absence of the CSA module, the network fails to align partials 3 and 5 between support and query samples when the target's pose varies. These findings underscore the CSA module's dual functionality explicitly aligning cross-pose semantics to mitigate geometric distortions and enabling PGI to focus on spatially consistent partial features for explainable fine-grained reasoning. The cascaded degradation in both alignment precision and discriminative power confirms that CSA can enhance model robustness against viewpoint variations, a key challenge in few-shot fine-grained remote sensing classification.

F. Hyperparameter Tuning

1) Partial Num (PN): In the PGI Module, we introduced an adjustable hyperparameter h, named PN, which plays a key role in learning fine-grained partials with discriminative semantics through attention mechanisms. To evaluate the impact of this hyperparameter on IPM-Net's performance in few-shot fine-grained classification tasks on remote sensing images, as well as its influence on the interpretability analysis provided by PGI, we designed experiments to optimize the parameter.

We conducted experiments on the FGSC-23 and FGSCR-42 datasets, applying both 5-way 1-shot and 5-way 5-shot tasks to test IPM-Net. Partial Num was adjusted to values of 1, 3, 5, and 10, with corresponding classification accuracy results shown in Table VII. The experiments exhibited that the classification ability of the network has a trend of first increasing and then decreasing as the parameter was varied from 1 to 10. This special variation pattern indicates the impact of partial number on IPM-Net. Specifically, when Partial Num is set to 1, the network achieves a baseline value of 52.95% for classification accuracy on FGSCR-42 dataset. As the parameter increased to 5, the network classification accuracy peaked at 57.83%, demonstrating that the network's partial-level representation achieves optimal consistency and alignment with true features. However, when Partial Num further increases to 10, the accuracy drops to 54.39%, reflecting limited feature representation ability due to overly fine-grained and dispersed attention partials.

Besides, we utilized PGI for a qualitative analysis of Partial Num, visualizing the evolution of partial forms as the parameter varied from 3 to 10. The result shows an increasing partial number resulted in larger and more concentrated partials, whereas smaller values of the parameter led to more dispersed and finer partials. Taking the Akizuki-class destroyer as an example in scene with Partial Num set to 3, as shown in Fig. 11, it has been observed that in order to acquire sufficient information from a smaller number of partials, each partial covers a larger area and is relatively distinct, to the extent that partial 2 focuses entirely on areas outside the hull target. Meanwhile, the scope attended to by partials 1 and 3 is broader and lacks fine granularity. When Partial Num is set to 5, the partials are not only capable of focusing on features from different regions, but each partial corresponds to a smaller area. This indicates that the network has learned partials with a certain level of fine granularity, which can effectively recognize subtle features with distinct semantics, such as the island superstructure, runway, and parked aircraft.

This also addresses the question raised in Section IV-D Interpretability Analysis. Appropriately incorporating useful background information can enhance the network's classification accuracy. Setting an appropriate partial number allows the network to learn relatively refined and useful background information. This approach is more effective than both introducing all background information and not introducing any background information at all. When Partial Num is increased to 10, the focused features become more minute. Due to the excessive number of partials, there is even a case of redundancy. Only a minority of partials have learned the most useful information for classification, while the majority of the partials are not assigned high weights in the final decision-making process. This results in the overall attention map being

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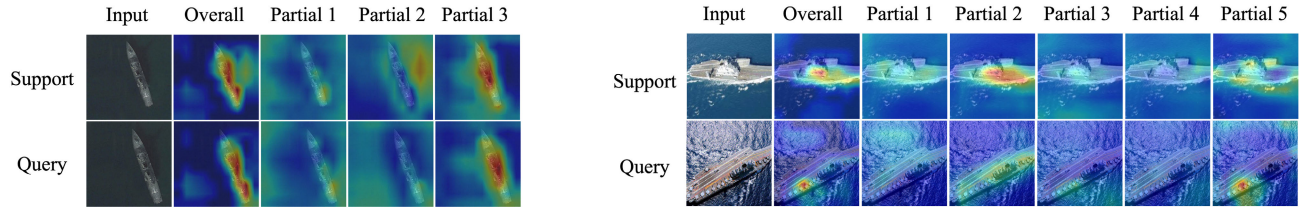


Fig. 11. Visualization of partials in IPM-Net with varying PN parameters. Partial-level features utilized by the network during classification for learning and matching. The impact of different numbers of partials is shown to analyze the influence of this parameter on network performance. (a) Sample pairs of akizuki-class destroyers with partial number set to 3. (b) Sample pairs of container ship with partial number set to 5.

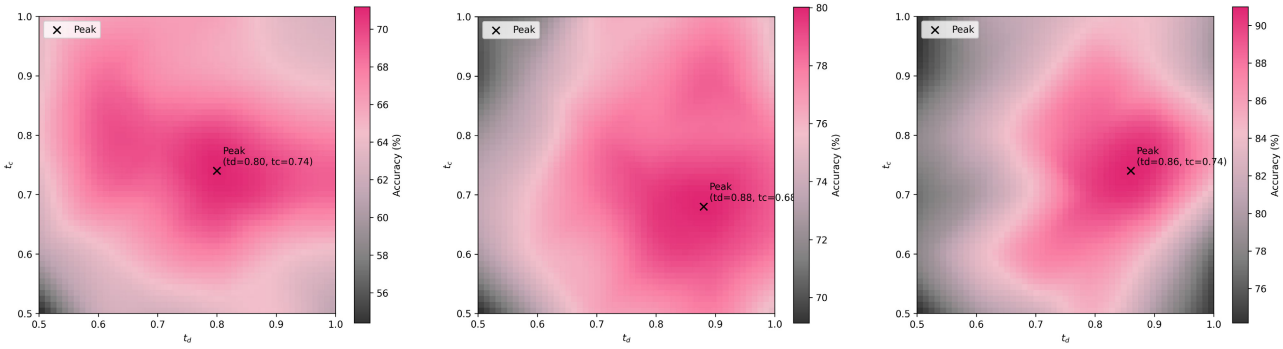


Fig. 12. Influence of the threshold parameters t_d and t_c on model accuracy across different datasets under the 5-shot setting. Where (a) FGVC-Aircraft dataset, (b) FGSCR-42 dataset, and (c) CUB-200-bb dataset, respectively.

composed primarily of the attention from a few dominant partials.

We conclude that the Partial Num parameter significantly influences the network's attention and feature aggregation. Lower values guide the network to broader, more coherent feature representations, while higher values enhance its ability to capture fine-grained details, albeit at the cost of fragmented attention and reduced coherence. Striking a balance is critical: excessively low values oversimplify features, hindering subtle distinctions, whereas excessively high values compromise coherence. Our qualitative and quantitative analyses suggest that a mid-range value, such as 5, achieves optimal Interpretability and robust classification performance across datasets in few-shot settings.

2) Threshold in the RaR Module: To further examine the behavior of the RaR Module, we conducted controlled experiments to analyze the influence of the threshold parameters t_c and t_d. These two hyperparameters operate at different stages with distinct objectives: t_c governs foreground segmentation in the *Refine* stage, while t_d controls salience suppression in the *Rediscover* stage.

a) Threshold mechanisms: Both thresholds follow percentile-based adaptive normalization [(23)], with distinct roles. The **Refine threshold** t_c retains pixels above the t_cth percentile, segmenting the foreground from the background. A higher t_c effectively eliminates background interference and focuses on salient features, while a lower t_c risks including background clutter. The **Rediscover threshold** t_d suppresses pixels exceeding the t_dth percentile, forcing the model to explore secondary features. Higher t_d suppresses only dominant features, encouraging broader exploration, while lower t_d may suppress useful discriminative cues, hindering exploration.

b) Performance analysis: A grid search of both thresholds was conducted across three datasets (FGVC-Aircraft, FGSCR-42, and CUB-200-2011) from 0.5 to 1.0 (step 0.1). Performance improves as t_c increases from 0.5 to 0.7, where an optimal balance is achieved by retaining the top 30% salient pixels, effectively isolating the foreground. Beyond $t_c = 0.7$, further increases lead to over-cropping, degrading performance due to the loss of essential features. For t_d, performance improves with increasing values from 0.5 to 0.8, where suppression of the top 20% dominant features achieves the best results, allowing exploration of secondary discriminative regions. Heatmaps generated from the 6×6 accuracy grid revealed distinct performance trends, with peak response regions identified around $t_c = 0.7$ and $t_d = 0.8$, as shown in Fig. 12. The visualization results confirm the consistent behavior of the RaR module across all datasets.

c) Selection guidelines: Based on the sensitivity analysis, we recommend selecting t_c in the range of $[0.65, 0.75]$. This range ensures sufficient background suppression while preserving the foreground, with lower values (0.65–0.68) suitable for complex backgrounds and higher values (0.72–0.75) for cleaner datasets. For t_d, we suggest a range of $[0.75, 0.85]$, with higher values (0.80–0.85) favored for fine-grained tasks and lower values (0.75–0.80) for coarser tasks. The optimal

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TABLE VIII
MODEL PARAMETERS, FLOATING POINT OPERATIONS, AND THE ACCURACY ON FGSCR-42 DATASET ARE COMPARED

Method	FLOPs(G)	Params(M)	Accuracy(%)
CrossTransformer [9]	3.017	0.1	43.74
TDM [8]	3.009	**0.2**	43.74
FRN [7]	3.018	0.1	45.47
RIFRN [10]	3.495	**0.2**	53.74
Ours	**3.520**	**0.2**	**57.83**

configuration is $t_c = 0.7$ and $t_d = 0.8$, which maximizes performance across all datasets.

G. Discussion

1) Analyze of Model Complexity: To evaluate the model complexity of **IPM-Net**, both the number of parameters and floating-point operations per second (FLOPs) were analyzed on the FGSCR-42 dataset under the 5-way 1-shot setting. We compared IPM-Net with several representative approaches, including CrossTransformer, TDM, FRN, and RIFRN, to assess its computational efficiency. The results, shown in Table VIII, reveal that although incorporating the PGI Module and the Cross-pose Semantic Alignment (CSA) Module leads to a moderate increase in parameters, the overall growth in model complexity remains limited. Specifically, the FLOPs of IPM-Net increase by only 0.025 compared to RIFRN, and the parameter count remains unchanged relative to RIFRN but rises by merely **0.1 million** when compared to CIFRN. Despite this slight increase, IPM-Net achieves a substantial improvement in classification accuracy—showing a gain of **12.11%** over FRN and **3.84%** over RIFRN. These results demonstrate that the proposed architecture achieves high performance without imposing a significant computational burden.

2) Analyze of Few-Shot Model Interpretability: From a broader perspective, embedding interpretability as a functional prior enables the model to focus on causally relevant and semantically consistent cues. This mechanism explains why IPM-Net remains effective on both natural and remote-sensing datasets, the model learns discriminative feature and transferable partial–relation rather than domain-specific textures or contexts. The interpretable architecture thus bridges visual reasoning from support data to unseen sample, offering a paradigm for few-shot learning. This paradigm suggests a direction of leveraging interpretable feature games as an intrinsic learning principle to achieve causally grounded, domain-robust, and human-understandable visual intelligence.

V. CONCLUSION

In this article, we proposed IPM-Net, a new framework for few-shot fine-grained classification in remote sensing imagery. The proposed design integrates three key components: the PGI Module, the Cross-pose Semantic Alignment (CSA) Module, and the RaR Module. PGI formulates partial-level feature extraction as a cooperative game, enabling the network to derive semantically meaningful and discriminative partials. CSA further enhances representation consistency by aligning these partials across varying poses, while RaR improves robustness through foreground refinement and complementary feature recovery. Evaluations on five fine-grained benchmarks, including both remote sensing and natural image datasets, demonstrate that IPM-Net achieves SOTA performance on all remote sensing tasks and exhibits strong transferability to natural image scenarios. The method maintains high interpretability, supported by quantitative metrics and visually coherent partial-level explanations. Ablation studies confirm the necessity of each module, and hyperparameter analysis identifies effective operational ranges for reliable deployment. Overall, the results indicate that incorporating a game-theoretic interpretability mechanism into the feature learning process may enhance the model's ability to transfer knowledge from base classes to novel classes. By guiding the network to reason over partial-level feature interactions, the approach strengthens the capture of fine-grained and semantically discriminative features that are essential for few-shot generalization. Thus, introducing cooperative games may be a feasible direction for developing interpretable few-shot fine-grained classification methods. Future work will further investigate the potential of this paradigm.

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